

# Seabin Lee

Seoul, South Korea | [sebin-lee-sg.github.io/Seabin-main](https://sebin-lee-sg.github.io/Seabin-main) | [123sebin@naver.com](mailto:123sebin@naver.com) | [Google Scholar](#) | [GitHub](#)



## PROFESSIONAL SUMMARY

Robotics researcher specializing in large-scale **Multi-Robot Task Allocation (MRTA)** for **Multi-Robot Systems**, spanning both multi-mobile-robot and multi-manipulator platforms. Also working on **Multi-Agent Path Finding (MAPF)** and **Human-Robot Interaction (HRI)** for scalable coordination and supervision, and on building simulation environments that combine real-time interaction, visualization, and robot behavior modeling.

## EDUCATION

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| <b>Sogang University</b><br><i>Ph.D. in Electronic Engineering (Advisor: Prof. Changjoo Nam)</i> | Seoul, South Korea<br>Mar 2024 – Present  |
| <b>Sogang University</b><br><i>M.S. in Electronic Engineering (Advisor: Prof. Changjoo Nam)</i>  | Seoul, South Korea<br>Mar 2022 – Feb 2024 |
| <b>Sogang University</b><br><i>B.S. in Electronic Engineering</i>                                | Seoul, South Korea<br>Mar 2016 – Feb 2022 |

## PUBLICATIONS

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| <b>Enhancing Situation Awareness in Multi-Robot Supervision: The Role of Adaptive Interfaces for Multiple Displays</b><br><i>Int. Journal of Human-Computer Interaction (IJHCI)   S. Park, S. Lee, S. Park, C. Nam   Published</i> | 2026 |
| <b>Very Large-Scale Multi-Robot Task Allocation in Challenging Environments via Robot Redistribution</b><br><i>Robotics and Autonomous Systems (RAS)   S. Lee, J. Sim, C. Nam   Published   <a href="#">Code</a></i>               | 2025 |
| <b>Coordination of Multiple Mobile Manipulators for Ordered Sorting of Cluttered Objects</b><br><i>IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)   J. Ahn, S. Lee, C. Nam   Published</i>                           | 2023 |

## PREPRINTS & UNDER REVIEW

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| <b>HEART: Coordination of Heterogeneous Expert Agents for Physically Grounded Robotic Task Planning</b><br><i>arXiv:2606.25404   J. Lee*, S. Lee*, W. Lee, N. Kim, M. Kang, C. Nam   Under Review</i> | 2026 |
| <b>Reinforcement Learning for Robotic Insertion of Flexible Cables in Industrial Settings</b><br><i>arXiv:2509.13731   J. Park, S. Lee, C. Park, W. Lee, C. Nam   Under Review</i>                    | 2025 |

## DOMESTIC CONFERENCES

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| <b>Task Allocation Considering Movement Paths of Multiple Autonomous Vehicles (in Korean)</b><br><i>Summer Conf. of the Korea ICT Platform Society   S. Lee, C. Nam</i> | 2023 |
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## RESEARCH IN PROGRESS

|                                                                                                                                                                                                                                  |         |
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| <b>Human-Factor-Aware Multi-Robot Supervision Task Allocation</b><br><i>Multi-robot supervision, human factors</i><br>– Allocating tasks across a robot fleet while accounting for the human supervisor’s workload and attention | Ongoing |
| <b>Task Allocation and Path Planning for Multi-Manipulator Systems</b><br><i>Multi-manipulator coordination</i><br>– Joint task allocation and collision-free motion planning for multiple manipulators sharing a workspace      | Ongoing |
| <b>Online MAPD in Multi-Floor Buildings with Elevators</b><br><i>Multi-Agent Pickup and Delivery</i><br>– Online task assignment and path planning for robots that move between floors using shared elevators                    | Ongoing |

## EXPERIENCE

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| <b>Graduate Researcher, AI Robotics Lab (AIRLAB), Sogang University</b> | Mar 2022 – Present |
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## SKILLS

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| <b>Languages:</b> Python, C++, C# <b>Simulation &amp; Robotics:</b> Isaac Sim, cuRobo, Unity, Blender <b>Tools:</b> Linux, Git |
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